

Final Exam

Macroeconomics II

Hui-Jun Chen

Problem 1 (Questions 1–20): Measurement, IS–LM, AD–AS

- ① Units. If CPI increases from 180 to 189 over a year, inflation is closest to ____
- (A) 2%
 - (B) 5%
 - (C) 9%
 - (D) –5%
- ② Approximation. For small changes, $\log(P_t/P_{t-1})$ is approximately ____
- (A) $P_t - P_{t-1}$
 - (B) $(P_t - P_{t-1})/P_{t-1}$
 - (C) $(P_t - P_{t-1})/P_t$
 - (D) P_{t-1}/P_t
- ③ Fisher (numbers). If $i = 1.5\%$ and $\mathbb{E}\pi = 2.0\%$, the ex-ante real rate is ____
- (A) 3.5%
 - (B) 0.5%
 - (C) –0.5%
 - (D) 2.0%
- ④ ZLB. If $i = 0$ and expected inflation falls from 1% to –1%, the real rate changes by ____

- (A) -2 pp
- (B) 0 pp
- (C) $+2$ pp
- (D) $+1$ pp

⑤ Debt valuation view. The price level tends to rise when expected future primary surpluses _____

- (A) increase
- (B) fall
- (C) stay constant
- (D) equal current taxes

⑥ Okun. If $u - u^n = -0.4x$ and $x = -3\%$, then $u - u^n$ is _____

- (A) -1.2%
- (B) $+1.2\%$
- (C) -0.75%
- (D) $+0.75\%$

⑦ Money market numerics. If $M/P = 12$, $k = 0.8$, $h = 4$, and $Y = 20$, then i equals _____

- (A) 1
- (B) 2
- (C) 3
- (D) 4

⑧ IS numerics. If $Y = 200 - 50r$ and r falls by 0.01 , then $\Delta Y =$ _____

- (A) -0.5
- (B) $+0.5$
- (C) $+5$
- (D) -5

⑨ AD slope (IS–LM). With M fixed, $P \uparrow$ leads to $Y \downarrow$ because _____

- (A) real balances rise
- (B) real balances fall and i rises
- (C) IS shifts left mechanically
- (D) Y^{FE} rises

⑩ AD–AS numerics. AD: $y = 50 - 2p$, SRAS: $p = 1 + 0.5(y - 40)$. Equilibrium output y is _____

- (A) 40
- (B) 42
- (C) 44
- (D) 46

⑪ Using the previous AD–AS system, the equilibrium price level p is _____

- (A) 1
- (B) 2
- (C) 3
- (D) 4

⑫ Diagnosis. A positive demand shock in AD–AS typically causes _____

(A) $y \uparrow, p \uparrow$

(B) $y \downarrow, p \uparrow$

(C) $y \uparrow, p \downarrow$

(D) $y \downarrow, p \downarrow$

⑬ **Diagnosis.** A negative supply shock in AD–AS typically causes _____

(A) $y \uparrow, p \uparrow$

(B) $y \downarrow, p \uparrow$

(C) $y \uparrow, p \downarrow$

(D) $y \downarrow, p \downarrow$

⑭ **Medium run.** Output returns toward y^{FE} mainly because _____

(A) AD shifts back automatically

(B) expected prices/wages adjust (SRAS shifts)

(C) money supply is constant

(D) productivity always rises

⑮ **IS–LM:** a cut in lump-sum taxes (holding G fixed) is typically modeled as _____

(A) IS shifts right

(B) LM shifts right

(C) IS shifts left

(D) LM shifts left

⑯ **IS–LM:** a decrease in nominal money M (holding P fixed) shifts _____

(A) LM right/down

(B) LM left/up

(C) IS right

(D) IS left

- ①⑦ Liquidity trap logic. If the money-market-clearing formula implies $i^* < 0$ and there is a ZLB, then the equilibrium i is _____
- (A) i^*
 - (B) 0
 - (C) $|i^*|$
 - (D) undetermined always
- ①⑧ One-time jump. If P rises by 6% once and then stays constant, inflation is _____
- (A) 6% forever
 - (B) 6% once then 0%
 - (C) 0% forever
 - (D) -6% once then 0%
- ①⑨ Anchored expectations means _____
- (A) $\mathbb{E}\pi$ stays near target after temporary shocks
 - (B) $\pi = 0$ always
 - (C) $x = 0$ always
 - (D) i never changes
- ②⑦ IS–MP–PC uses which three blocks? _____
- (A) IS, LM, Phillips
 - (B) IS, policy rule, Phillips
 - (C) AD, money demand, LRAS
 - (D) budget constraint, money demand, Phillips
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Problem 2 (Questions 21–40): IS–MP–PC, Taylor principle, NK

- (21) In Whelan's IS–MP derivation, the key term is $(\beta_\pi - 1)$ because the real-rate gap contains _____
- (A) $i - \pi$
 - (B) $i + \pi$
 - (C) $\pi - i$
 - (D) i/π
- (22) Taylor principle. $\beta_\pi > 1$ ensures that when inflation rises, the real rate _____
- (A) falls
 - (B) rises
 - (C) is unchanged
 - (D) must be zero
- (23) If $\theta = \frac{1}{1 + \alpha\gamma(\beta_\pi - 1)}$ and $\beta_\pi > 1$, then _____
- (A) $\theta > 1$
 - (B) $\theta = 1$
 - (C) $0 < \theta < 1$
 - (D) $\theta < 0$
- (24) Numerics. If $\alpha = 1.2$, $\gamma = 0.4$, $\beta_\pi = 2$, then θ is closest to _____
- (A) 0.68
 - (B) 0.32
 - (C) 1.68
 - (D) 0

- (25) A tougher central bank corresponds to _____
- (A) lower β_π
 - (B) higher β_π
 - (C) higher γ
 - (D) lower α
- (26) With a tougher CB, a cost-push shock tends to produce _____
- (A) larger inflation rise, smaller output fall
 - (B) smaller inflation rise, larger output fall
 - (C) smaller inflation rise, smaller output fall
 - (D) no changes
- (27) Explosive region in the TaylorRule discussion: if $1 - \frac{1}{\alpha\gamma} < \beta_\pi < 1$, then _____
- (A) IS–MP slopes down and is stable
 - (B) IS–MP slopes up and can be unstable under adaptive expectations
 - (C) inflation is pinned at target
 - (D) Phillips curve becomes vertical
- (28) Boundary. If $\alpha\gamma = 0.5$, then $1 - \frac{1}{\alpha\gamma} =$ _____
- (A) 0.5
 - (B) -1
 - (C) 1
 - (D) 2
- (29) Two-period NK IS (calculation). With $x_1 = 0$, $\sigma = 1$, $r_0^n = 1\%$, $i_0 = 4\%$, and $\mathbb{E}_0\pi_1 = 2\%$, x_0 is _____

- (A) +1%
- (B) 0%
- (C) -1%
- (D) -3%

③ Forward-looking NKPC is _____

- (A) $\pi_0 = \pi_{-1} + \kappa x_0 + u_0$
- (B) $\pi_0 = \beta \mathbb{E}_0 \pi_1 + \kappa x_0 + u_0$
- (C) $\pi_0 = \kappa x_0$ always
- (D) $\pi_0 = \beta \pi_0 + \kappa x_0$

③ NKPC numerics. If $\beta = 0.99$, $\mathbb{E}_0 \pi_1 = 1\%$, $\kappa = 0.3$, $x_0 = 2\%$, $u_0 = 0$, then π_0 is _____

- (A) 1.59%
- (B) 1.29%
- (C) 0.99%
- (D) 0.60%

③ Taylor rule numerics. If $i_0 = \bar{i} + \phi_\pi \pi_0$ with $\bar{i} = 1\%$, $\phi_\pi = 1.5$, and $\pi_0 = 3\%$, then i_0 is _____

- (A) 3%
- (B) 4.5%
- (C) 5.5%
- (D) 6%

③ Two-period Fisher numerics. If $i_0 = 0\%$ and $\mathbb{E}_0 \pi_1 = 2\%$, then r_0 is _____

- (A) 2%
- (B) 0%
- (C) -2%
- (D) -4%

③4 Two-period IS sign. Holding $\mathbb{E}_0\pi_1$ fixed, a decrease in i_0 implies _____

- (A) x_0 decreases
- (B) x_0 increases
- (C) x_0 unchanged
- (D) π_0 falls one-for-one

③5 In NKPC, increasing u_0 is best interpreted as _____

- (A) a right shift in IS
- (B) an upward shift in inflation for given x_0 and expectations
- (C) a fall in r_0^n
- (D) a decrease in κ

③6 Two-period closure. If $x_1 = 0$ and $\pi_1 = 0$ are imposed, then at $t = 0$ _____

- (A) there are no expectations terms
- (B) the system is purely backward-looking
- (C) policy becomes irrelevant
- (D) the LM curve returns

③7 Determinacy in basic NK. A standard sufficient condition is _____

- (A) $\phi_\pi < 1$
- (B) $\phi_\pi > 1$
- (C) $\kappa = 0$
- (D) $\sigma = 0$

- ③⑧ Two-period NK IS (calculation). Suppose $x_1 = 0$, $\sigma = 1$, $r_0^n = 1\%$, $i_0 = 3\%$, and $\mathbb{E}_0\pi_1 = 0\%$. Then x_0 is _____
- (A) +2%
 - (B) -2%
 - (C) -1%
 - (D) 0%
- ③⑨ IS-MP (calculation). If $\alpha = 1.5$ and $\beta_\pi = 0.8$, then $-\alpha(\beta_\pi - 1)$ equals _____
- (A) -0.3
 - (B) +0.3
 - (C) -1.2
 - (D) +1.2
- ④⑩ IS-MP slope (sign). If $\beta_\pi = 0.6$ and $\alpha > 0$, then IS-MP in (y, π) space is _____
- (A) downward sloping
 - (B) upward sloping
 - (C) vertical
 - (D) flat